Preview: y2007_q04 (v1)

Digit sums: find the next/previous number with the same digit sum

Part A: Each question is worth 3 points

1. The numbers 34 and 142 have the same sum of digits (3 + 4 = 7 and 1 + 4 + 2 = 7). What is the first number greater than 967 whose sum of digits is the same as for 967?

A. 968 ***B.*** 969 ***C.*** 972 ***D.*** 976 ***E.*** 977

2. The numbers 34 and 142 have the same sum of digits (3 + 4 = 7 and 1 + 4 + 2 = 7). What is the first number greater than 7386 whose sum of digits is the same as for 7386?

A. 7387 ***B.*** 7390 ***C.*** 7391 ***D.*** 7395 ***E.*** 7458

3. Which number has the same sum of digits as 131?

A. 13 ***B.*** 14 ***C.*** 15 ***D.*** 220 ***E.*** 400

4. Which number has the same sum of digits as 7915?

A. 7627 ***B.*** 7754 ***C.*** 7925 ***D.*** 7961 ***E.*** 8067

Part B: Each question is worth 4 points

5. What is the smallest number whose digits add up to 15?

A. 55 ***B.*** 59 ***C.*** 68 ***D.*** 69 ***E.*** 70

6. What is the smallest number whose digits add up to 30?

A. 3008 ***B.*** 3040 ***C.*** 3479 ***D.*** 3566 ***E.*** 3999

7. What is the biggest number smaller than 721 whose sum of digits is the same as for 721?

A. 712 ***B.*** 711 ***C.*** 703 ***D.*** 649 ***E.*** 589

8. What is the biggest number smaller than 3834 whose sum of digits is the same as for 3834?

A. 3825 ***B.*** 3833 ***C.*** 3843 ***D.*** 4275 ***E.*** 4505

Part C: Each question is worth 5 points

9. How many two-digit numbers have digits that add up to 13?

A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 12

10. How many three-digit numbers have digits that add up to 5?

A. 15 ***B.*** 14 ***C.*** 13 ***D.*** 6 ***E.*** 5

Answer Key

1.	**D**	**2.**	**D**	**3.**	**B**	**4.**	**A**	**5.**	**D**	**6.**	**E**
7.	**A**	**8.**	**A**	**9.**	**D**	**10.**	**A**				

Solutions

1. D (976) — 9 + 6 + 7 = 22. Going up from 967, the first number with digit sum 22 is 976 (9 + 7 + 6 = 22). [2007 Q4 · next_same_sum]

2. D (7395) — 7 + 3 + 8 + 6 = 24. Going up from 7386, the first number with digit sum 24 is 7395 (7 + 3 + 9 + 5 = 24). [2007 Q4 · next_same_sum]

3. B (14) — 1 + 3 + 1 = 5 and 1 + 4 = 5. The others add up to 4 or 6. [2007 Q4 · which_same_sum]

4. A (7627) — 7 + 9 + 1 + 5 = 22 and 7 + 6 + 2 + 7 = 22. The others add up to 21 or 23. [2007 Q4 · which_same_sum]

5. D (69) — Use as few digits as possible: as many 9s as possible, with the smallest digit in front. 69: 6 + 9 = 15. [2007 Q4 · smallest_with_sum]

6. E (3999) — Use as few digits as possible: as many 9s as possible, with the smallest digit in front. 3999: 3 + 9 + 9 + 9 = 30. [2007 Q4 · smallest_with_sum]

7. A (712) — 7 + 2 + 1 = 10. Counting down from 721, the first number with digit sum 10 is 712. [2007 Q4 · before_same_sum]

8. A (3825) — 3 + 8 + 3 + 4 = 18. Counting down from 3834, the first number with digit sum 18 is 3825. [2007 Q4 · before_same_sum]

9. D (6) — List them by the first digit: 49, 58, 67, 76, 85, 94. That is 6. [2007 Q4 · count_with_sum]

10. A (15) — Count by the first digit: 5 + 4 + 3 + 2 + 1 = 15. [2007 Q4 · count_with_sum]